

Smart Campus Digital Twin

Full Technical Architecture, Frontend/Backend Systems, Physical Campus Research & Deployment Specification

Project Status: Production Ready & Verified Live	Live URL: smart-campus-digital-twin-eight.vercel.app
Physical Campus: Anekal Master Campus (60 Acres)	3D Assets: Blender 4.3 (.blend) + WebGL GLTF (.glb)

1. Executive Summary & Vision

The Alliance University Smart Campus Digital Twin is a cyber-physical spatial management platform developed as a flagship capstone engineering project for the Department of Computer Science & Engineering (Class of 2026). It integrates real-time Three.js WebGL simulation, 1,000+ streaming IoT sensor feeds, 480 kWp rooftop solar microgrid distribution, predictive load forecasting, autonomous spatial allocation, and direct facility actuator overrides across the 60-acre Anekal campus in Bengaluru.

2. Physical Campus Research & Geographic Grounding

Unlike generic 3D dashboards, this platform is mathematically grounded in the real-world physical layout of Alliance University's central campus on Chandapura-Anekal Main Road, Bengaluru (Coordinates: 12.845° N, 77.684° E). The spatial hierarchy and road network were reconstructed from Google Maps satellite imagery and aerial surveys:

- **Campus Gateway:** Entrance aligned alongside Senior City, feeding into the primary arterial boulevard.
- **Admin Block (ADMIN):** Signature ground-floor neoclassical building on the right with open rotunda dome, colonnaded entry portico, and registrar offices.
- **Parking Quad:** Multi-bay vehicle surface parking situated immediately on the left across from the Admin Block.
- **Divided Central Boulevard:** Dual-carriageway road stretching ~100 meters inwards, lined with architectural trees and landscaped median.
- **Central Library (LIB):** 4-floor, 50,000 sq ft circular rotunda dome building on the left side of the boulevard.
- **Engineering Block (ACED):** 7-floor flagship engineering facility following the left-hand turn, housing CSE, AI/ML, and robotics labs.
- **Athletic Complex:** Open grounds opposite B.Tech containing cricket practice nets, volleyball court, and multi-sport football ground.
- **Hostels & Microgrid:** Student residential halls with perimeter green buffers, 480 kWp solar PV, 1.2 MWh BESS battery, and STP water recycling.

3. Frontend Architecture & Technology Stack

The frontend is engineered with React 19, TypeScript, and Vite, leveraging Tailwind CSS Glassmorphism (Wispr Flow) and Three.js hardware acceleration for 60 FPS in-browser rendering:

Component / Layer	Technology	Architectural Role
Core Application	React 19, TypeScript, Vite	SPA state coordination, reactive telemetry hooks, HMR
WebGL 3D Engine	Three.js, @react-three/fiber	60 FPS hardware-accelerated canvas, dynamic sun lighting
3D Camera & Controls	@react-three/drei (OrbitControls)	Architectural waypoints, bounded orbital navigation

Design System	Tailwind CSS v3, CSS Variables	Wispr Flow glassmorphism, backdrop blurs, 5 live themes
Digital Twin Panel	DigitalTwinPanel.tsx	Mode switcher (Thermal/Occupancy/Energy), GLB loader
Actuator Dock	ManualControlDock.tsx	Floating macOS dock with auto-hide on scroll (>80px)
Microgrid Sankey	SankeyEnergyFlow.tsx	Real-time power routing between PV, BESS, and campus
Predictive Charting	EnergyTrendChart.tsx	24-hour demand forecasting vs. solar generation curves
Space Allocation	RoomTimeline.tsx, MiniMapCard	Classroom scheduling and ghost-booking reclamation
Brand & Favicon Suite	PWA Manifest + 7 Icons	Official Alliance University crest, mobile PWA icons

4. Backend, Simulation & Data Services

- **Physics-Coupled IoT Telemetry:** Simulates 1,000+ IoT beacons. Solar irradiance is mathematically derived from sun elevation angle (peaks at 480 kWp at 12:30 PM solar noon). Thermal absorption and HVAC loads scale dynamically with occupancy.
- **Autonomous Campus Evolution Agent:** Background event loop (campusEvolutionAgent.ts) that scans spatial feeds, identifies empty conditioned rooms, and recommends automated energy reallocations.
- **Headless Blender Python Automation:** Custom Python script (perfect_alliance_campus.py) executed in Blender 4.3 to procedurally construct, texture, light, and export both .blend and .glb assets.
- **Edge CDN Delivery:** Deployed on Vercel with Brotli compression and single-file bundling for sub-second load times worldwide.

5. Manual Facility Actuators & Controls

- **Atmosphere & Solar Time:** Scrub time from 06:00 to 22:00 to simulate daylight, golden hour, dusk, and nighttime starfields.
- **BESS Battery Storage:** Toggle 1.2 MWh storage bank between Auto, Grid Charging, and 310 kW Peak-Shaving Discharge.
- **STP Water Circulation:** Manually activate or stop the 421 kL main wastewater treatment lift pump.
- **Chiller Setpoint:** Regulate campus chilled water temperatures (19°C to 26°C) with live impact on predicted grid draw.
- **Floor Plate Explosion:** Separate floor plates (0% to 100%) to inspect interior floor plans without displacing building structure.

6. Key Engineering Challenges Solved

- **OrbitControls Wheel Trapping:** Unbound default Three.js canvas mouse-wheel zoom so trackpad/wheel gestures scroll the webpage naturally. Added dedicated HUD Zoom In/Out (+ / -) buttons and Shift-key wheel zoom.
- **Collision-Free Dock Layout:** Re-anchored the floating scroll prompt symmetrically at bottom-24 right-8 with auto-hide on scroll (>80px) to prevent overlapping with dock cards.
- **Fixed Navbar Offsetting:** Added scroll-padding-top: 5rem and scroll-mt-20 so jumping to dashboard sections never hides headings.
- **Branding & Favicon Suite:** Integrated complete 7-item favicon suite with official Alliance University crest and PWA manifest.